I. Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset

A. Data type of all columns in the "customers" table

```
select column_name,
       data_type
from `target-487907.Target.INFORMATION_SCHEMA.COLUMNS`
where table_name = 'customers'
order by ordinal_position ;
```

Row	column_name	data_type
1	customer_id	STRING
2	customer_unique_id	STRING
3	customer_zip_code_prefix	INT64
4	customer_city	STRING
5	customer_state	STRING

data type looks correct for every data.

B. Get the time range between which the orders were placed.

```
select min(order_purchase_timestamp) as first_order,
       max(order_purchase_timestamp) as last_order
from `Target.orders` ;
```

Query results

+ Create conversation

Job information | Results | Visualization | JSON | Execution deta

Row	first_order	last_order
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC

C. Count the Cities & States of customers who ordered during the given period.

```
select count(distinct c.customer_city) as unique_city_count,
       count(distinct c.customer_state) as unique_state_count
from `Target.orders` o
inner join `Target.customers` c
on o.customer_id = c.customer_id ;
```

Query results

+ Create conversation

Job information | Results | Visualization | JSON | Execution detai

Row	unique_city_count	unique_state_cou...
1	4119	27

Geographic Coverage Insight

Target has customer transactions in **4,119 cities** out of 5570 cities, which represents approximately **73.94% of all Brazilian municipalities**.

At the state level, Target operates in **27 out of 27 administrative regions**, achieving **100% state-level coverage**.

This suggests that Target is well-established throughout Brazil, with a large distribution in both states and cities. To attain complete nationwide coverage, it may yet be possible to extend into the remaining cities.

II. In-depth Exploration:

A. Is there a growing trend in the no. of orders placed over the past years?

```
select extract(year from order_purchase_timestamp) order_year,
       count(order_id) as total_orders
from `Target.orders`
group by 1
order by 1 ;
```

Row	order_year	total_orders
1	2016	329
2	2017	45101
3	2018	54011

The company recorded a growth rate of **12,608.51% between 2016 and 2017**. However, this figure is significantly inflated because the first order was placed in September 2016, meaning only four months of sales data are available for that year. As a result, the 2016–2017 comparison does not reflect a true full-year growth trend.

A more meaningful comparison is between **2017 and 2018**, where the company achieved a healthy growth rate of **19.75%**, indicating steady and sustainable expansion.

B. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

```
select extract(month from order_purchase_timestamp) as order_month,
       count(order_id) as total_orders
from `Target.orders`
group by 1
order by 1 ;
```

Row	order_month	total_orders
1	1	8069
2	2	8508
3	3	9893
4	4	9343
5	5	10573
6	6	9412
7	7	10318
8	8	10843
9	9	4305
10	10	4959

The number of orders changes during the year. Orders are higher between **May and August**, with **August having the highest number (10,843 orders)**. **September and October** have the **lowest number of orders**, which shows a slowdown during those months. Overall, the middle months of the year perform better compared to the later months.

C. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

```
select case
         when extract(hour from order_purchase_timestamp) between  0 and 6 then 'Dawn'
         when extract(hour from order_purchase_timestamp) between  7 and 12 then
'Morning'
         when extract(hour from order_purchase_timestamp) between 13 and 18 then
'Afternoon'
         else 'Night' end as time_period,
       count(order_id) as total_orders
from `Target.orders`
group by 1
order by 2 desc ;
```

Row	time_period	total_orders
1	Afternoon	38135
2	Night	28331
3	Morning	27733
4	Dawn	5242

Most customers place their orders in the **afternoon**, with 38,135 orders, making it the busiest time of the day. Night and morning have almost similar order volumes. Very few orders are placed during dawn hours.

This shows that customers are more active during the afternoon, possibly during work breaks or leisure time.

III. Evolution of E-commerce orders in the Brazil region:

A. Get the month on month no. of orders placed in each state.

```
select c.customer_state,
       extract(year from o.order_purchase_timestamp) as year_orders,
       extract(month from o.order_purchase_timestamp) as month_orders,
       count(o.order_id) as total_orders
from `Target.customers` c
inner join `Target.orders` o
on c.customer_id = o.customer_id
group by 1, 2, 3
order by 1, 2, 3 ;
```

Row	customer_state	year_orders	month_orders	total_orders
1	AC	2017	1	2
2	AC	2017	2	3
3	AC	2017	3	2
4	AC	2017	4	5
5	AC	2017	5	8
6	AC	2017	6	4
7	AC	2017	7	5
8	AC	2017	8	4
9	AC	2017	9	5
10	AC	2017	10	6

The month-on-month analysis across states shows consistent growth from 2017 to 2018. Economically stronger states such as São Paulo (SP), Rio de Janeiro (RJ), and Minas Gerais (MG) contribute to the highest order volumes, while smaller states record significantly lower activity. This indicates that Target's performance is concentrated in major economic regions, though growth is visible across most states.

B. How are the customers distributed across all the states?

```sql
select customer_state,
       count(distinct customer_id) as number_of_customers
from `Target.customers`
group by 1
order by 1;
```

Row	customer_state	number_of_customers
1	AC	81
2	AL	413
3	AM	148
4	AP	68
5	BA	3380
6	CE	1336
7	DF	2140
8	ES	2033
9	GO	2020
10	MA	747

Customer distribution is **uneven** across states. São Paulo (SP) has the **highest number of customers (41,746)**, followed by Rio de Janeiro (RJ) and Minas Gerais (MG). These states contribute a **large portion** of the total customer base.

On the other hand, smaller states such as Roraima (RR), Amapá (AP), and Acre (AC) have **very few customers**.

This shows that customer activity is **concentrated** in economically stronger and more populated states.

IV. Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

A. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).

```
with all_orders as(
select extract(year from o.order_purchase_timestamp)as order_year,
       extract(month from o.order_purchase_timestamp) as order_month,
       p.payment_value
from `Target.orders` o
inner join `Target.payments` p
on o.order_id = p.order_id
where extract(year from o.order_purchase_timestamp) in (2017, 2018)
and extract(month from o.order_purchase_timestamp) between 1 and 8
),

order_cost as(
select order_year,
       sum(payment_value) as cost_of_orders,
       lag(sum(payment_value)) over(order by order_year) as previous_cost_of_orders
from all_orders
group by 1
)

select order_year,
       order_cost.cost_of_orders,
       round(100 * (order_cost.cost_of_orders - previous_cost_of_orders) /
previous_cost_of_orders, 2) as increase_in_cost_of_orders
from order_cost ;
```

Row	order_year	cost_of_orders	increase_in_cost_...
1	2017	3669022.119999...	*null*
2	2018	8694733.840000...	136.98

The total cost of orders increased from **3.67 million in Jan–Aug 2017 to 8.69 million in Jan–Aug 2018**, representing a **136.98% increase**. This shows that revenue more than doubled within one year, indicating strong business expansion.

B. Calculate the Total & Average value of order price for each state.

```
select c.customer_state,
       round(sum(p.payment_value),2) as total_order_price,
       round(sum(p.payment_value) / count(distinct o.order_id),2) as avg_order_price
from `Target.orders` o
inner join `Target.payments` p
on o.order_id = p.order_id

inner join `Target.customers` c
on o.customer_id = c.customer_id

group by 1
order by 1 ;
```

Row	customer_state	total_order_price	avg_order_price
1	AC	19680.62	242.97
2	AL	96962.06	234.77
3	AM	27966.93	188.97
4	AP	16262.8	239.16
5	BA	616645.82	182.44
6	CE	279464.03	209.18
7	DF	355141.08	165.95
8	ES	325967.55	160.34
9	GO	350092.31	173.31
10	MA	152523.02	204.18

Revenue is highly concentrated in **São Paulo (SP)**, which generates nearly **6 million** in total order value, far higher than any other state. RJ and MG also contribute significantly, while smaller states generate comparatively lower revenue.

C. Calculate the Total & Average value of order freight for each state.

```sql
with freight_value as(

select o.order_id,
       c.customer_state,
       sum(oi.freight_value) as order_freight_value
from `Target.orders` o
inner join `Target.order_items` oi
on o.order_id = oi.order_id

inner join `Target.customers` c
on o.customer_id = c.customer_id
group by 1, 2
)

select customer_state,
       round(sum(freight_value.order_freight_value),2) as total_freight_value,
       round(avg(freight_value.order_freight_value),2) as avg_freight_value
from freight_value
group by 1
order by 1 ;
```

Row	customer_state	total_freight_value	avg_freight_value
1	AC	3686.75	45.52
2	AL	15914.59	38.72
3	AM	5478.89	37.27
4	AP	2788.5	41.01
5	BA	100156.68	29.83
6	CE	48351.59	36.44
7	DF	50625.5	23.82
8	ES	49764.6	24.58
9	GO	53114.98	26.46
10	MA	31523.77	42.6

While total freight cost is highest in major states like SP and RJ due to high order volume, the average freight per order is lowest in these states. In contrast, smaller and remote states such as RR, PB, and RO show higher average freight costs. This suggests that **logistics efficiency is stronger in high-demand regions and relatively expensive in low-density regions.**

V. Analysis based on sales, freight and delivery time.

A. Find the no. of days taken to deliver each order from the order's purchase date as delivery time. Also, calculate the difference (in days) between the estimated & actual delivery date of an order. Do this in a single query.

```
select order_id,
       timestamp_diff(order_delivered_customer_date, order_purchase_timestamp, day) as
delivery_time,
       timestamp_diff(order_estimated_delivery_date, order_delivered_customer_date,
day) as day_diff
from `Target.orders`
where lower(order_status) = 'delivered'
and order_delivered_customer_date is not null ;
```

Row	order_id	delivery_time	day_diff
1	bfbd0f9bdef84302105ad712db...	54	-36
2	98974b076b01553d49ee64679...	43	6
3	c4b41c36dd589e901f6879f25a...	36	14
4	d2292ff2201e74c5db154d1b7a...	29	20
5	95e01270fcbae986342340010...	30	19
6	ed8c7b1b3eb256c70ce0c7423...	44	5
7	5cc475c7c03290048eb2e742c...	68	-18
8	6b3ee7697a02619a0ace2b3f0a...	47	2
9	3b2ca3293a7ce539ea2379d70...	43	7
10	b2f92b2f7047cd8b35580d629d...	43	7

The delivery_time column represents the number of days taken from purchase to actual delivery.

The day_diffcolumn shows the difference between estimated and actual delivery date.

B. Find out the top 5 states with the highest & lowest average freight value.

```
with freight_cost as(
select c.customer_state,
       round(sum(oi.freight_value) / count(distinct oi.order_id),2) as
avg_freight_value
from `Target.orders` o
inner join `Target.order_items` oi
on o.order_id = oi.order_id

inner join `Target.customers` c
on o.customer_id = c.customer_id
group by 1
),

cost_ranking as(
select customer_state,
       freight_cost.avg_freight_value,
       dense_rank() over(order by freight_cost.avg_freight_value desc) as top_5,
       dense_rank() over(order by freight_cost.avg_freight_value) as bottom_5
from freight_cost
)

select customer_state as states,
       avg_freight_value as avg_freight_cost,
       case
         when top_5 <= 5 then 'Top 5'
         when bottom_5 <= 5 then 'Bottom 5' end as category
from cost_ranking
where top_5 <= 5
or bottom_5 <= 5
order by avg_freight_cost desc ;
```

Row	states	avg_freight_cost	category
1	RR	48.59	Top 5
2	PB	48.35	Top 5
3	RO	46.22	Top 5
4	AC	45.52	Top 5
5	PI	43.04	Top 5
6	RJ	23.95	Bottom 5
7	DF	23.82	Bottom 5
8	PR	23.58	Bottom 5
9	MG	23.46	Bottom 5
10	SP	17.37	Bottom 5

The query calculates the average freight value for each state and identifies the top 5 states with the highest and lowest average value.

C. Find out the top 5 states with the highest & lowest average delivery time.

```
with delivery_time_per_state as(

select c.customer_state,
       o.order_id,
       timestamp_diff(o.order_delivered_customer_date, o.order_purchase_timestamp,
day) as delivery_time_days
from `Target.customers` c
inner join `Target.orders` o
on o.customer_id = c.customer_id
where lower(o.order_status) = 'delivered' and
      o.order_delivered_customer_date is not null
),

state_ranking as(

select customer_state,
       avg(delivery_time_days) as avg_delivery_time,
       dense_rank() over(order by avg(delivery_time_days) desc) as top_5,
       dense_rank() over(order by avg(delivery_time_days)) as bottom_5
from delivery_time_per_state
group by 1
)

select customer_state,
       avg_delivery_time,
       case
        when top_5 <= 5 then 'Top 5 states'
        when bottom_5 <= 5 then 'Bottom 5 states' end as category
from state_ranking
where top_5 <= 5 or
      bottom_5 <= 5
order by avg_delivery_time desc ;
```

Row	customer_state	avg_delivery_time	category
1	RR	28.97560975609...	Top 5 states
2	AP	26.73134328358...	Top 5 states
3	AM	25.98620689655...	Top 5 states
4	AL	24.04030226700...	Top 5 states
5	PA	23.31606765327...	Top 5 states
6	SC	14.47518330513...	Bottom 5 states
7	DF	12.50913461538...	Bottom 5 states
8	MG	11.54218777523...	Bottom 5 states
9	PR	11.52671135486...	Bottom 5 states
10	SP	8.298093544722...	Bottom 5 states

The query calculates the average delivery time (in days) for each state and identifies the top 5 states with the highest and lowest average delivery durations.

D. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

```
with delivery_day_diff as(

select c.customer_state,
       o.order_id,
       timestamp_diff(order_estimated_delivery_date, order_delivered_customer_date,
day) as day_difference
from `Target.orders` o
inner join `Target.customers` c
on o.customer_id = c.customer_id
where lower(order_status) = 'delivered'
and order_delivered_customer_date is not null
),

top_5_states as(

select customer_state,
       avg(day_difference) as avg_day_difference,
       dense_rank() over(order by avg(day_difference)desc) as top_5
from delivery_day_diff
group by customer_state
)

select customer_state,
       case when top_5 <= 5 then 'Top 5 states' end as category
from top_5_states
where top_5 <= 5 ;
```

Row	customer_state	category
1	AC	Top 5 states
2	RO	Top 5 states
3	AP	Top 5 states
4	AM	Top 5 states
5	RR	Top 5 states

The query returns to the **top 5 states where delivery was fastest compared to the estimated delivery date**, based on the highest average positive difference between estimated and actual delivery dates.

VI. Analysis based on the payments

A. Find the month on month no. of orders placed using different payment types.

```
select extract(year from o.order_purchase_timestamp) as order_year,
       extract(month from o.order_purchase_timestamp) as order_month,
       p.payment_type,
       count(distinct o.order_id) as total_orders
from `Target.orders` o
inner join `Target.payments` p
on o.order_id = p.order_id
group by 1, 2, 3
order by 1, 2, 3 ;
```

Row	order_year	order_month	payment_type	total_orders
1	2016	9	credit_card	3
2	2016	10	UPI	63
3	2016	10	credit_card	253
4	2016	10	debit_card	2
5	2016	10	voucher	11
6	2016	12	credit_card	1
7	2017	1	UPI	197
8	2017	1	credit_card	582
9	2017	1	debit_card	9
10	2017	1	voucher	33

The **month-on-month** analysis shows that **credit card is the dominant payment method** across all months, **followed by UPI**. **Debit card and voucher payments account for a smaller share** of total transactions. **A significant spike in order volume is observed in November 2017** across all payment types, indicating strong seasonal demand.

B. Find the no. of orders placed on the basis of the payment installments that have been paid.

```
select p.payment_installments,
       count(distinct o.order_id) as total_orders
from `Target.orders` o
inner join `Target.payments` p
on o.order_id = p.order_id
where lower(o.order_status) = 'delivered' and
      p.payment_installments >= 1
group by 1
order by 1 ;
```

Row	payment_installm...	total_orders
1	1	47586
2	2	12052
3	3	10147
4	4	6882
5	5	5090
6	6	3800
7	7	1560
8	8	4122
9	9	618
10	10	5137

Many customers prefer paying in a **single installment (47,586 orders)**. Installment options between **2 and 6** are moderately used, while **higher installment plans show significantly lower usage**, indicating **limited preference for long-term EMI options**.

- ## Final Strategic Recommendations

1. Develop Regional Fulfillment Centers in High Freight & High Delivery-Time States

States such as **RR, AP, AM, and RO** show both higher average freight costs and longer delivery times.

Target should consider localized warehousing or micro-fulfillment hubs in northern and remote regions to reduce shipping distance, lower freight costs, and improve delivery speed.

2. Strengthen Logistics Efficiency in Low-Density Regions

High-demand states like **SP, RJ, and MG** demonstrate lower average freight and faster delivery due to economies of scale.
Target should replicate this operational efficiency model in underperforming regions through better route optimization and regional carrier partnerships.

3. Maximize Revenue in High-Contribution States

Revenue is highly concentrated in **SP, RJ, and MG**.
Target can introduce:

- Premium product bundles
- Loyalty programs
- Personalized promotions

to increase customer lifetime value in these high-performing markets.

4. Optimize Payment Strategy Based on Customer Behavior

Since **47,586 orders are paid in a single installment**, customers show a strong preference for upfront payments.
Instead of focusing on long-term EMI plans (15+ installments), Target should prioritize short-term installment offers (2–6 months), where moderate demand exists.

5. Capitalize on Seasonal Demand Peaks

A significant spike in orders was observed in **November 2017**, especially via credit card transactions.
Target should:

- Strengthen inventory planning
- Increase marketing budget before peak months
- Offer strategic card-based promotions

To maximize seasonal revenue.

6. Data-Driven Regional Expansion Strategy

Although Target has presence in **4,119 cities (73.94% coverage)** and full state coverage, customer concentration and revenue are heavily skewed toward SP, RJ, and MG.

Target should:

- Identify high-population cities within underperforming states
- Launch targeted marketing campaigns in low-customer states
- Offer region-specific discounts to stimulate demand
- Partner with local sellers to improve assortment relevance

This ensures:

- Balanced national growth
- Reduced revenue dependency on a few major states
- Long-term sustainable expansion

Conclusion

Overall, the analysis indicates that Target has achieved strong nationwide presence and consistent revenue growth, particularly between 2017 and 2018. Customer and revenue concentration is heavily skewed toward major economic states such as SP, RJ, and MG, while logistics inefficiencies are visible in low-density regions. Payment behavior shows a strong preference for single-installment transactions, and seasonal spikes suggest clear demand cycles. With targeted logistics optimization and regional growth strategies, Target can further improve operational efficiency and expand sustainable revenue across Brazil.